SEMICONDUCTORS IN NANODEVICES

Undergraduate Course Project Report

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Course Project
2026

Abstract

This report covers the role of semiconductors in nanodevices, starting from the basic physics of energy bands and charge carriers up to fabrication methods and current applications. The main semiconductor materials used at the nanoscale are compared based on properties such as band gap, electron mobility, and flexibility. The steps used to fabricate a nanoscale transistor are described, along with the main challenges faced during fabrication, such as precision, defects, cost, and scalability. The report also covers current applications of semiconductor nanomaterials in electronics, photonics, energy conversion, and sensing, and closes with a discussion of the environmental impact of semiconductor manufacturing and ongoing research in the field.

I. INTRODUCTION

A. Background

Semiconductors are used in almost every electronic device we use daily, including mobile phones, TVs, and laptops. Over the last few years, semiconductor nanodevices have become one of the fastest growing areas in electrical and nanotechnology engineering, since reducing the size of semiconductor structures to the nanoscale changes how they behave. At this scale, materials such as silicon (Si), gallium arsenide (GaAs), and transition metal dichalcogenides (TMDCs) show quantum mechanical effects, including quantum confinement and tunable band gaps, that are not present in bulk materials.

These effects allow engineers to control the electrical and optical properties of a material more precisely, which opens the door to applications in flexible electronics, nanomedicine, and neuromorphic computing, in addition to the standard use in transistors and integrated circuits.

B. Problem Description

Understanding how semiconductor nanodevices work requires combining knowledge from several areas: the physics of energy bands, the properties of different semiconductor materials, and the fabrication processes used to build nanoscale structures. Each of these areas affects the final performance of a device, and choosing the wrong material or fabrication method for a given application can lead to poor performance, high cost, or a device that cannot be manufactured at scale.

This project addresses this by reviewing each of these areas and connecting them, so the reasoning behind material selection and fabrication choices in real nanodevices becomes clear.

C. Objectives

The objectives of this project are:

- Explain the fundamental physics that governs semiconductor behavior, including energy bands, charge carriers, and doping.
- Compare the main semiconductor materials used in nanodevices and identify which material fits which application.
- Describe the fabrication methods used to build nanoscale semiconductor devices, using the nanoscale transistor as the main example.
- Review current and emerging applications of semiconductor nanomaterials.
- Discuss the environmental impact of semiconductor manufacturing and the direction of ongoing research.

II. FUNDAMENTAL PHYSICS OF SEMICONDUCTORS

A. Energy Bands and Conductivity

Materials can be classified into conductors, semiconductors, and insulators based on the size of their band gap, which is the energy required for an electron to move from the valence band to the conduction band. Conductors have loosely bound valence electrons and effectively no band gap, insulators have tightly bound valence electrons and a very large band gap, and semiconductors fall in between with a moderate band gap. This moderate gap is what allows

the conductivity of a semiconductor to be controlled, which is the property that makes them useful in electronic devices.

The relation between band gap and conductivity is straightforward: a larger band gap means more energy is required for an electron to reach the conduction band, so the material conducts less easily. A material with a very large band gap is called a wide-band-gap semiconductor.

B. Charge Carriers

Charge carriers are the particles responsible for carrying electric current inside a semiconductor. There are two types: electrons, which are negatively charged, and holes, which represent the absence of an electron and behave as if they carry a positive charge of the same magnitude. Both carriers move under an applied electric field and together determine the conduction behavior of the material.

C. Doping and Extrinsic Semiconductors

A pure semiconductor, free of impurities, is called an intrinsic semiconductor. In practice, most semiconductors used in devices are extrinsic, meaning they have been doped with an impurity to change their electrical properties. Doping is the process of adding an impurity to a semiconductor to increase its ability to conduct electricity.

The type of impurity added determines the resulting semiconductor type. Adding trivalent atoms produces a P-type semiconductor, which has an excess of positive charge carriers (holes), while adding pentavalent atoms produces an N-type semiconductor, which has an excess of negative charge carriers (electrons). N-type and P-type regions are combined to build most semiconductor devices, including diodes and transistors.

III. SEMICONDUCTOR MATERIALS FOR NANODEVICES

A. Material Overview

Choosing the right material is critical at the nanoscale, because device performance depends heavily on properties such as electron mobility, band gap, flexibility, and chemical stability. Table I summarizes the four main materials covered in this project, together with their typical use and main limitation.

TABLE I. SEMICONDUCTOR MATERIALS AND THEIR LIMITATIONS

Material	Typical Use	Limitation
Silicon (Si)	Microchips, solar cells (most widely used material in electronics)	Lower electron mobility compared to newer materials
Gallium Arsenide (GaAs)	High-speed electronics, LEDs, microwave frequency devices	More expensive and more brittle than silicon
Graphene	Advanced electronics, sensors, flexible displays, batteries	No natural band gap, difficult to switch off current
Molybdenum Disulfide (MoS2)	Next-generation transistors, photodetectors, flexible electronics	Still in early development, less mature than silicon

B. Material Selection Criteria

Each nanodevice application has different requirements, so the same material is not optimal for every case. Mobility is a measure of how easily an electron or hole moves inside the semiconductor for a given temperature and doping level, and it is one of the main properties considered when selecting a material. Flexibility, meaning the ability of a material to bend without breaking, is another important factor, especially for wearable and foldable electronics. Materials such as carbon nanotubes and graphene combine high flexibility with high strength, which is a relatively rare combination.

Table II shows the band gap values of two common materials as an example of how this property affects device suitability.

TABLE II. BAND GAP COMPARISON

Material	Band Gap (eV)	Note
Silicon	~1.1	Suitable for transistors and integrated circuits
Germanium	~0.66	Conducts more easily than silicon

C. Applications by Material

Based on the properties above, different materials are chosen for different nanodevice applications. Table III lists the material choices identified for four common application areas.

TABLE III. MATERIAL CHOICES BY APPLICATION

Application	Material Choices
Transistors (processors, memory, logic)	Silicon (Si) - industry standard, low cost; MoS_2 - 2D material with natural band gap and good scaling
Photodetectors / solar cells	Gallium Arsenide (GaAs) - high conversion efficiency; Quantum dots (e.g., CdSe) - tunable band gap
Flexible electronics (wearables, foldable screens)	Graphene - strong, flexible, high conductivity; Organic semiconductors - low cost, printable
Sensors (gas sensors, biosensors)	Graphene / MoS_2 - large surface area, sensitive to molecules; Carbon nanotubes - high surface-to-volume ratio

IV. FABRICATION METHODS

A. Photolithography

Photolithography is one of the main industrial methods used to fabricate nanoscale semiconductor structures. It works by using light to transfer a pattern from a photomask onto a light-sensitive photoresist coated on the wafer. The process follows four main steps: coating the wafer with photoresist, exposing the photoresist to UV light through a mask, developing the photoresist to reveal the pattern, and then etching or depositing material before removing the remaining photoresist.

Photolithography offers high throughput and is cost-effective for large-scale production, which is why it remains the standard method in the industry. Its main limitation is resolution: standard deep ultraviolet (DUV) lithography is limited by diffraction to about 193 nm, while

extreme ultraviolet (EUV) lithography can reach resolutions of about 2 nm, but at a much higher cost.

B. Nanoscale Transistor Fabrication Process

As an example of how these methods are combined in practice, Table IV outlines the main steps used to fabricate a nanoscale transistor such as a FinFET.

TABLE IV. MAIN STEPS IN FABRICATING A NANOSCALE TRANSISTOR

Step	Stage	Description
1	Substrate Preparation	Start with a cleaned and polished silicon wafer
2	Fin Formation	Use photolithography and etching to create 3D silicon fins
3	Gate Dielectric Deposition	Deposit a high-k dielectric (e.g., HfO_2) using atomic layer deposition (ALD)
4	Gate Electrode Formation	Deposit a metal gate (e.g., TiN) using CVD or sputtering
5	Source/Drain Doping	Implant dopants to form source and drain regions
6	Spacer Formation	Deposit insulating spacers (e.g., SiN) to isolate the gate from source/drain
7	Interconnects	Deposit metal layers (e.g., copper) for electrical connections
8	Testing and Packaging	Final electrical testing and packaging of the device

C. Fabrication Challenges

Fabricating devices at the nanoscale introduces several practical challenges:

- Precision: achieving features below 10 nm requires extremely accurate tools such as EUV or e-beam lithography, which are complex and costly.
- Defects: atomic-scale defects, such as vacancies and dislocations, can degrade device performance, so contamination control is critical.
- Cost: advanced tools, such as EUV lithography machines, cost hundreds of millions of dollars, and developing new processes is expensive.
- Scalability: balancing high precision with high manufacturing throughput remains difficult.

V. APPLICATIONS AND ONGOING RESEARCH

A. Nanoelectronics, Photonics and Sensing

Semiconductor nanomaterials have useful properties such as narrow and intensive emission spectra, continuous absorption bands, and good chemical and photobleaching stability, which make them attractive for a wide range of applications. Because the transport of electrons, holes, phonons, and photons in these materials is strongly affected by their size and geometry, semiconductor nanomaterials are used in nanoelectronics, nanophotonics, energy conversion, miniaturized sensors and imaging devices, solar cells, catalysis, detectors, and biomedicine.

One example is the use of semiconductor photocatalysis for hydrogen production. Hydrogen is considered a promising alternative fuel because it is pollution-free and can be produced from renewable energy sources, and photocatalysis is expected to contribute to both environmental treatment and renewable energy generation.

B. Silicon-Based and Organic Optoelectronic Devices

Silicon remains the foundation of modern microelectronics, and silicon nanostructures are considered a key building block for future nano- and micro-electronic devices. As CMOS circuit dimensions continue to shrink toward 10 nm and below, silicon-based nanotechnology is expected to provide a path to lower cost and higher efficiency devices.

Beyond silicon, organic optoelectronic materials are also used to build nanodevices such as organic light-emitting diodes (OLEDs), organic photovoltaic cells (OPVs), and organic thin-film transistors (OTFTs). These devices are attractive because of their solid-state operation, self-emission, full-color capability, and flexibility, and are already used in flat-panel and flexible displays and solid-state lighting.

Semiconductor quantum dots (QDs) are another area of interest. Because their size is reduced to the nanometer scale in all three dimensions, quantum dots have discrete, atom-like energy levels that depend on their size. This allows their light absorption and emission color to be tuned, which is useful for display and imaging applications.

C. Ongoing Research

Research on semiconductor nanodevices is still active. As an example, three research teams at the UCLA Samueli School of Engineering were awarded a combined $3.38 million in grants by the National Science Foundation through its Future of Semiconductors program, led by professors Kang Wang, Benjamin Williams, and Danijela Cabric. One of these projects, led by Williams' group, received a three-year, $1.96 million grant to develop hybrid electronic-photonic semiconductor devices operating at frequencies above 1 terahertz (THz), which is beyond what current silicon-foundry semiconductors can reliably generate and detect.

VI. DISCUSSION

A. Advantages

Semiconductor nanodevices offer tunable electrical and optical properties that cannot be achieved with bulk materials, since band gap, mobility, and light emission can all be adjusted through material choice, doping, and nanostructure size. This tunability is what enables applications ranging from high-speed transistors to flexible sensors and tunable-color quantum dots.

B. Limitations and Challenges

The main limitations come from fabrication rather than the materials themselves. Producing reliable nanoscale features requires extremely precise and expensive tools, and even small atomic-scale defects can significantly affect device performance. Some promising materials, such as MoS_2, are still not mature enough for large-scale production compared to silicon.

C. Environmental Considerations

Semiconductor manufacturing is resource-intensive and consumes large amounts of energy, which raises environmental concerns as demand for electronic devices keeps growing. Wide-band-gap materials such as silicon carbide (SiC) and gallium nitride (GaN) can improve energy efficiency in power devices, reducing energy losses compared to standard silicon

devices. On the manufacturing side, chemical recycling processes are being developed to recover materials from semiconductor waste, and ultrasonic cleaning is being used to reduce the amount of water and chemicals needed during fabrication.

VII. CONCLUSION

Semiconductors are essential building blocks of nanodevices, enabling high-performance electronics, sensors, and optoelectronic components. Their tunable properties, particularly band gap and carrier mobility, make them suitable for transistors, LEDs, and quantum devices. Fabricating these devices at the nanoscale is still limited by precision, defect control, and cost, but ongoing advances in nanomaterials such as quantum dots, nanowires, and 2D materials are expected to lead to faster, smaller, and more efficient devices in the future, with applications extending into computing, healthcare, and energy.

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